Sahaj Bhadja

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Summary

Software engineer with 2+ years building **business applications**, REST APIs, and data-backed workflows. Strong in **SQL**, **Python**, **JavaScript/TypeScript**, CI/CD, testing, and modernization of operational software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
M.S. in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
B.Tech. in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Eng., OS, OOP, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant platform services on Azure SQL and Blob Storage for 500+ users, optimizing queries to cut dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant workflows for 1K+ background jobs with state transitions and failure isolation, improving throughput 3x while supporting dependable releases, validation, and debugging.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Flask services, REST APIs, and instructor-facing React workflows handling 10K+ daily requests, integrating secure file ingestion, grading pipelines, and polished user-facing logic.
- Designed concurrent batch-processing for 1K+ uploaded files using multithreading and AWS RDS-backed services, reducing turnaround time by 60% while maintaining stable APIs in production.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform software with microservices, Jenkins, PyTest, and Mockito, improving release reliability through 20+ automated test suites across Windows, Linux, and macOS.
- Optimized API and ETL workflows with AWS Glue, S3, and SQL-backed services, improving response times by 30% and pipeline efficiency by 68% for production operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed full-stack workflow software using REST APIs, WebSockets, and responsive interfaces, supporting 2M+ synchronized records while improving operational visibility, coordination, and usability.
- Automated Docker-based build and deployment workflows with Jenkins CI/CD, cutting release time by 50% and improving troubleshooting speed, environment consistency, and delivery reliability.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built backend APIs for a cross-platform booking application, covering reservations, scheduling, status updates, and payment workflows across 50K+ records through reliable Flask services.
- Mapped 12+ integration points and implemented payment and synchronization features with Flask, Firebase, and webhooks, reducing manual payment handling by 70%.

Skills

Backend & Data: SQL Server, Azure SQL, PostgreSQL, MySQL, REST APIs, Flask, FastAPI, Node.js, microservices, OOP, debugging, unit/integration testing
Frontend & Apps: JavaScript, TypeScript, HTML5, CSS3, React, responsive UI, line-of-business workflows, application modernization
Cloud & Delivery: AWS, Azure, GCP, Docker, Kubernetes, Jenkins, CI/CD, Git, Bash, Agile/Scrum, security-minded development, API integration

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management through a unified Java and TypeScript app.

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Built a healthcare operations platform with scheduling, alerts, and dashboards, reducing administrative workload by 30% through structured TypeScript services, monitoring, and automation.

CPU-scheduling Simulator Live | *C++, React, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with C++ logic, metrics, and Gantt visualization across 1K+ simulated process combinations.